A MORPHISM OF GROUPOIDS BIJECTIVE ON ORBITS AND ON ISOTROPY GROUPS IS AN EQUIVALENCE

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Allowed standard prerequisites (editorial).

- The orbit set pi_0 of a groupoid and the isotropy groups of its objects. Recalled inside the counted proof at lines 1270-1276. Fixes the two invariants whose bijectivity the hypothesis asserts.
- A disjoint union of equivalences of groupoids is again an equivalence. Used inside the counted proof at lines 1276-1278. Turns the displayed decomposition of the groupoid into isotropy groups into the claimed equivalence.

Lemma 0.1. *Let $T\colon \mathcal{G} \to \mathcal{H}$ be a morphism of groupoids such that it induces a bijection between the sets of orbits and for each object $x \in \mathcal{G}$ an isomorphism of isotropy groups $\mathcal{G}_x^x \to \mathcal{H}_{T(x)}^{T(x)}$. Then T is an equivalence of groupoids.*

Date: doi:10.3842/SIGMA.2025.093; CC BY 4.0.

Proof. The assertion follows directly from the following commutative diagram:

$$\begin{array}{ccc} \bigsqcup_{[x]\in\pi_0(\mathcal{G})} \mathcal{G}_x^x & \xrightarrow{\cong} & \bigsqcup_{[x]\in\pi_0(\mathcal{H})} \mathcal{H}_{T(x)}^{T(x)} \\ \downarrow\simeq & & \downarrow\simeq \\ \mathcal{G} & \xrightarrow{T} & \mathcal{H}. \end{array}$$

Here, we denote $\pi_0(\mathcal{G})$ the set of orbits of $\mathcal{G}$ and we use that T induces a bijection between $\pi_0(\mathcal{G})$ and $\pi_0(\mathcal{H})$ and that a disjoint union of equivalences of groupoids is again an equivalence of groupoids. □